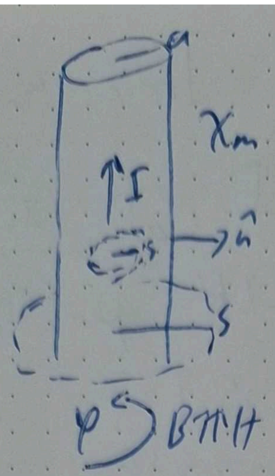
1)

13/1) a)



$s<a \Rightarrow \oint H \cdot dl = I_{fenc}$

$\Rightarrow 2\pi s |H| = \frac{\pi s^2 I}{\pi a^2}$

$\Rightarrow H = \frac{s I}{2\pi a^2} \varphi$

$s>a \quad \oint H \cdot dl = I_{fenc} \Rightarrow 2\pi s |H| = I$

$\Rightarrow H = \frac{I}{2\pi s} \varphi$

b) $s<a \quad B = \mu H = \mu_0 (H + \chi_m H) = \mu_0 (1+\chi_m) H$

$\Rightarrow \mu_0 (1+\chi_m) \frac{s I}{2\pi a^2} \varphi$

$s>a$ — " " —

$\Rightarrow B = \mu_0 \frac{I}{2\pi s} \varphi$

c) $J_b = \nabla \times M = \nabla \times \chi_m H = \frac{1}{s}\left(\frac{\partial}{\partial s}(s \chi_m H)\right) z$

$J_b = \frac{\chi_m I}{\pi a^2} z$

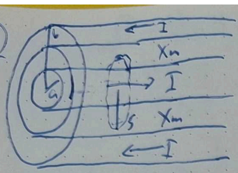
$K_B = M \times \hat{n} = \chi_m H \times \hat{n}$

$= -\chi_m H \cdot \sin\left(\frac{\pi}{2}\right) z$

$\Rightarrow K_B = -\frac{\chi_m I}{2\pi a} z$

2-4)

2)



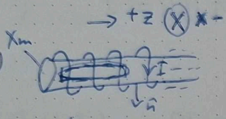
a) $\oint H \cdot dl = I_{f,enc}$ $a<s<b$

$2\pi s |H| = I \Rightarrow H = \frac{I}{2\pi s}\hat{\varphi}$

b) $B = \mu_0(1+X_m)H = \mu_0(1+X_m)\frac{I}{2\pi s}\hat{\varphi}$

c) $M = X_m H = \frac{X_m I}{2\pi s}\hat{\varphi}$

3) a)



$\oint H \cdot dl = I_{f,enc} \Rightarrow L|H| = nIL \Rightarrow H = nI\,\mathbf{z}$

b) $B = \mu_0(1+X_m)H = \mu_0(1+X_m)nI\,\mathbf{z}$

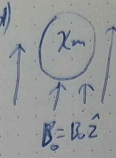
c) $M = X_m H = X_m nI\,\mathbf{z}$

d) $K_B = M \times \hat{n} = M \cdot \sin(90°)\,\mathbf{x} = -X_m nI\,\mathbf{x}$

e) Paramagnets: $X_m > 0 \Rightarrow K_B$ opposite to I

Diamagnets: $X_m < 0 \Rightarrow K_B$ same direction as I

4)



a) $H = \frac{1}{\mu_0}B - M$ & $\oint H \cdot dl = I_{f,enc} = 0$

$\Rightarrow B = \mu_0 M$ where M will allign with B if $X_m > 0$ (paramagnet) or oppose it $X_m < 0$ (diamagnet).

b) Inside field will be $B_{inside} = \mu_0 M$ wich depends on how much the material magnetizes from an external field.

c) The magnetic field will be slightly weaker or stronger inside depending on dia- or paramagnetic due to the induced dipole moment. But the net force will be zero so they don't move.

d) For ferromagnetic material the magnetization from B_0 is not linear and would change over time.